Project Brief

Project title	Agile Planning for Meals on Wheels Web Application
Module Name	Agile Project Management
Course Name	Higher Diploma in Software Engineering (E-Learning)
Project Start date	(Refer to the schedule)
Project Submission Date	(Refer to the schedule)

Purpose of this assignment

This Assignment is used for the following:

- Summative Assessment of students in the module, Agile Project Management, of the course, 'Higher Diploma in Software Engineering (E-Learning)'.

In this assignment, you have to implement project management using Scrum, plan projects, prioritize user stories, assign roles, execute sprints, and define sprints and success measures.

This assignment gives you an opportunity to demonstrate your capabilities in the following areas:

- Outline the Scrum Project
- Build the Scrum team & Implementation plan
- Create a backlog and prioritize user stories
- Execute a mock sprint, sprint review, and sprint retrospective using Scrum
- Create and manage your tasks and sprints using the ClickUp tool

Note: The tasks in this project are designed to test various agile project management concepts. In real applications, you might do it based on specific needs.

Table of Contents

Project Scenario 3
Project Requirements 4
Project Evidence 4
Project Guidelines 5
Project Technical Environment 5
Project Assumptions 5

Project Scenario

You are currently working as a full-stack web developer in a leading enterprise solution provider called Unity One Solutions. The company wanted to implement the project "Meals on Wheels". Your manager, Mr. David assigned you to design the software. You are required to work in a team to prepare a project plan using Scrum, prioritize user stories, assign roles, execute sprints, and define sprints and success measures.

MerryMeal is a charitable organization that prepares and delivers a hot noon meal to qualified adults living at home who are unable to cook for themselves or maintain their nutritional status due to age, disease, or disability. The service will be available Monday through Friday. Frozen meals will be provided to members who are not within a 10-kilometer radius of their outsourced kitchens and support over the weekend. MerryMeal has partnered with several food service providers across the country to provide the quickest delivery possible.

The Application should allow

1. Members and Care givers registration with their requirements
2. Partners and Volunteers registration with their details
3. Fund raising through Donors / Supporters
4. Menu Planning and Preparation
5. Meal Delivery Management of partners and riders
6. Food Safety Management
7. Reassessment of need evaluation, and,
8. Management Information System for effective management

Members and Caregivers Registration:
The application should allow members and caregivers to register with their requirements. Members should be able to provide their personal information, such as name, address, contact details, and dietary restrictions. Caregivers should be able to provide their personal information, contact details, and the details of the member they are caring for.

Partners and Volunteers Registration:
The application should allow partners and volunteers to register with their details. Partners should be able to provide their organization's details, such as name, address, contact details, and the types of services they can provide. Volunteers should be able to provide their personal information, contact details, and the types of services they can provide.

Fundraising through Donors/Supporters:
The application should have a feature for fundraising through donors and supporters. Donors should be able to donate money to MerryMeal using the application, and supporters should be able to pledge their support for the organization.

Menu Planning and Preparation:
The application should allow MerryMeal to plan and prepare menus for the meals they will be delivering. They should be able to add and remove dishes, set the quantity of food to be prepared, and track the availability of ingredients.

Meal Delivery Management of Partners and Riders:
The application should allow MerryMeal to manage the delivery of meals to members. They should be able to assign partners and riders to delivery routes, track the delivery status of each meal, and receive feedback from members on the quality of the service provided.

Food Safety Management:
The application should have a feature for food safety management. MerryMeal should be able to track the expiration dates of ingredients, track the preparation and cooking times of dishes, and maintain a record of food safety practices.

Reassessment of Need Evaluation:
The application should have a feature for reassessment of need evaluation. MerryMeal should be able to evaluate the needs of members periodically and adjust their services accordingly.

Management Information System for Effective Management:
The application should have a management information system that allows MerryMeal to monitor the performance of the organization. They should be able to track the number of meals delivered, the number of volunteers and partners registered, the amount of money donated, and other key performance indicators.

Project Requirements

You are required to submit a project management proposal and a clickup workspace to document the tasks below:

1. **Task 1: Build the Scrum Team**
   - 1.1 Recruit and onboard team members, assigning them to appropriate roles (Scrum Master, Product Owner, Developers, etc.).

2. **Task 2: Outline the Scrum Project & Implementation Plan**
   - 2.1 Draft a project scope document that outlines the overall objectives, deliverables, and timeline for the project.
   - 2.2 Identify key stakeholders and define roles.
   - 2.3 Identify the value this project creates for the business
   - 2.4 Define the outcomes expected if this project is successful
   - 2.5 Establish how project success will be measured (e.g., project completion on time, user satisfaction, etc.).
   - 2.6 Create a detailed plan outlining phases of the project (Requirement gathering, design and prototyping, development, testing, deployment, and maintenance), milestones, and dependencies.

3. **Task 3: Conduct a project kick-off meeting**
   - 3.1 Hold an initial meeting to introduce the team, discuss the implementation plan, and align on project goals.

4. **Task 4: Create user stories**
   - 4.1 Analyse of the scenario and business requirements and appropriate solution to meet the basic requirements together with extended enhancements that the business may require for future development.
   - 4.2 Work with stakeholders to document user requirements as user stories.

5. **Task 5: Prioritize User Stories and create a refined product backlog**
   - 5.1 Categorise the user stories into the Ralph Stacy Matrix
   - 5.2 Rank user stories in the backlog based on business value, urgency, and dependencies.
   - 5.3 Organize the prioritized user stories into a refined product backlog, ensuring alignment with the project's objectives

6. **Task 6: Plan the sprints, Sprint Review, and Sprint Retrospective Using Scrum**
   - 6.1 Select user stories from the backlog for the sprint, define sprint goals, break them into tasks, and estimate the effort required for each task.
   - 6.2 Define the discussion items and approach for the daily stand-ups
   - 6.3 Outline the approaches to be taken for Sprint review and retrospective
   - 6.4 Identify Definition of Done and any other criteria to measure the outcome of your implementation

7. **Task 7: Create and Manage Your Tasks and Sprints Using ClickUp**
   - 7.1 Configure ClickUp to support Scrum practices, including task statuses, custom fields, and automations.
   - 7.2 Break down user stories into manageable tasks and assign them to team members.
   - 7.3 Configure and evaluate sprint metrics like velocity, burndown, and completion rate to inform future sprints.

Project Evidence

Evidence checklist	Summary of evidence required by student which must be incorporated in the Project deliverables
Task 1	- Role assignment confirmations - Team structure and organization chart
Task 2	- Project Scope Document - Timeline and deliverable chart - Success Metrics, Key Value Areas and KPIs - Project Plan Document and Milestone Chart - Phase-wise breakdown with timelines - Dependency Matrix
Task 3	- Meeting Agenda and Minutes - Stakeholder Approval and Sign-off

Task 4	- User Stories - Stakeholder Review and Sign-off
Task 5	- Ralph Stacy Matrix - Ranking Criteria - Prioritized and refined Backlog
Task 6	- Sprint Planning Document - Selected User Stories - Task Breakdown Structure - Effort Estimation Report - Daily Stand-Up Agenda - Sprint Review and Retrospective Plan
Task 7	- Screenshots of Task Statuses, Custom Fields, and Automations - Sprint Metrics Dashboard - Velocity and Burndown Reports in ClickUp

Project Guidelines

You should follow the below guidelines while implementing the Project:

- Implement the project in the technical environment specified in the Project brief
- Follow the format specified for Project Report and Project Presentation
- The project report and presentation should be submitted at least 2 days before the date of Summative Assessment
- Present the Milestones in every mentoring Session and seek the Mentor's feedback and review, Incorporate the feedback in your project.
- Attach all project evidence for each milestone as part of your Project report

Project Technical Environment

The student should perform the project in the below environment. The student should have the following tools installed on his/her Windows 10 / Macintosh laptop with 8 GB RAM or equivalent.

- Click Up
- MS Office/OpenOffice tools

Project Assumptions

You can make following assumptions while implementing the project:

- This project is not for commercial use and hence not fine-tuned for performance, but it is desirable that you focus on performance issues and address them, if any